

need to be able to perform sensitivity and “what-if” analyses to determine the cost of trading under various market conditions. Managers need to be able to incorporate their own proprietary view of markets including their volatility and liquidity estimates, as well as their proprietary alpha estimates. Finally, managers do not want to be reliant upon what other brokers and vendors feel are appropriate values for the input variables, especially if these views differ from their own.

The current state of market impact models falls well short of the needs of portfolio managers. Broker models are often black box models, and most do not provide managers with sufficient transparency into the approach to allow the managers to evaluate or critique the results. Much of the reason why brokers keep these models so secretive and hidden is that they do not want the investment community to judge their models. To test this point, simply ask the broker salesperson to write the formulation of their market impact model, the definition of their input variables, and the model parameters. Then sit back and observe their responses. And if these parties do provide this information, try to duplicate their results for a few different samples of stocks.

Brokers will usually state numerous reasons why they are unable to provide their model to the client. They often claim that the model needs to be connected to a tic database, that the model is specific for their algorithms, and that the model uses a proprietary approach, or that the data cannot be redistributed. Regardless of the reason stated, investors should be extremely cautious of using any model or approach that is not amply described or transparent. These models have to be analyzed, tested, and verified.

To be fair, there may be some truth to why these models cannot be provided to the client. But it is still likely that the vendor is hesitant to provide the functional form because it may reveal that the model is not nearly as complex or sophisticated as claimed. Keeping the functional form of market impact models hidden from potential users makes it difficult for users to properly evaluate the model.

Suffice to say, that current industry market impact models have not evolved to a level needed by portfolio managers. A summary of these reasons, as stated in *Journal of Trading* (see Kissell, 2012), is as follows:

Current State of Vendor Market Impact Models

- Vendor models are black box approaches with no transparency. They do not provide the underlying formulas and often do not provide the complete set of input variables and explanatory factors used to